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DUNMAN HIGH SCHOOL

Preliminary Examination

Year 6

H2 CHEMISTRY

Paper 2 (Section A and B)

8872/01

21 September 2011

2 hours

Additional Materials: Writing Papers
Data Booklet

INSTRUCTIONS TO CANDIDATES

- 1 Write your **name** and **class** on all the work you hand in.
- 2 Write in dark blue or black pen on both sides of the paper.
- 3 You may use a soft pencil for any diagrams, graph or rough working.
- 4 The number of marks is given in brackets [] at the end of each question or part question.
- 5 You may use a calculator.
- 6 At the end of the examination, fasten all your work securely together.

SECTION A

Answer **all** questions.

SECTION B

Answer any **two** questions on separate answer paper.

Start each question on a fresh sheet of paper.

The total marks for this paper is 80 marks.

FOR EXAMINER'S USE	
Question	Marks
1	/14
2	/13
3	/13
Section A Subtotal	/40
4	/20
5	/20
6	/20
Section B Subtotal	/40
Paper 1 (MCQ)	/30
Total	/110
Grade	

Section A

Answer **all** questions in the space provided.

- 1 Naturally occurring magnesium has the following isotopic abundances:

Isotope	Abundance(%)	Atomic mass(amu)
^{24}Mg	78.99	23.98504
^{25}Mg	10.00	24.98584
^{26}Mg	11.01	25.98259

- (a) Define the term relative isotopic mass.

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[1]

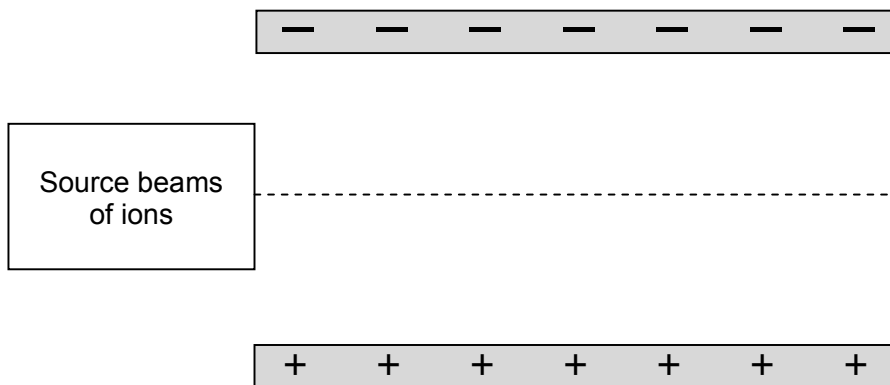
- (b) Calculate the average atomic mass of Mg.

[2]

- (c) Write an equation for the second ionisation energy of Mg.

[1]

- (d) A source beam of $^{24}\text{Mg}^{2+}$ and $^{127}\text{X}^{-}$ is passed through an electric field. Sketch the deflection paths of the ions in the diagram below. Show clearly their relative angles of deflection.



[2]

(e) Magnesium chloride, MgCl_2 , is an important coagulant used in the preparation of soy products. The *enthalpy change of formation* of MgCl_2 is -641 kJ mol^{-1} .

(i) Describe the type bonding formed by magnesium chloride and explain why magnesium chloride has a high boiling point.

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.....

(ii) Define, with the aid of an equation, the enthalpy change of formation of MgCl_2 .

.....

[3]

(f) The enthalpy change of formation of a hypothetical ionic solid, magnesium (I) chloride, MgCl , is -125 kJ mol^{-1} .

(i) Using the data given, calculate the enthalpy change of reaction, $\Delta H_{\text{reaction}}$ for the following reaction:



(ii) Hence comment on the stability of $\text{MgCl}(\text{s})$ relative to that of $\text{MgCl}_2(\text{s})$.

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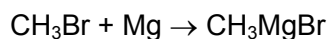
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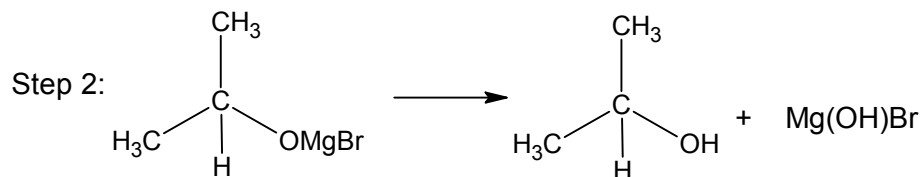
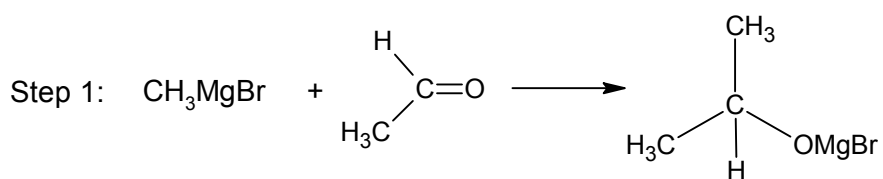
[3]

- (g) Magnesium metals can be used to make Grignard reagents, which are alkyl or aryl magnesium halides, often used for organic synthesis reactions.

In the preparation of a Grignard reagent, a halogenoalkane is added to magnesium solids in a flask containing dry ether as a solvent. The mixture is warmed over a water bath. The reaction is as follows:



This Grignard reagent can be used to synthesise propan-2-ol from ethanal. The reaction takes place in two steps as shown below:



- (i) State the type of reaction taking place in step 1.

- (ii) Suggest a suitable carbonyl compound that can be used to synthesise butan-2-ol using the Grignard reagent given.

[2]

[Total: 14]

- 2 Compound **W** has the molecular formula $C_5H_8O_3$. When **W** is refluxed with acidified potassium manganate(VII), only one organic product, **X**, is obtained. When **W** is warmed with aqueous sodium hydroxide and iodine, a yellow precipitate is formed. **W** has no reaction when 2,4–dinitrophenylhydrazine is added. One mole of **W** liberates one mole of hydrogen gas when treated with sodium metal.

(a) State the type(s) of reaction undergone when **W** is

(i) refluxed with acidified potassium manganate(VII)

.....

(ii) warmed with aqueous sodium hydroxide and iodine.

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[2]

(b) Name all the possible functional groups present in **W**.

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[1]

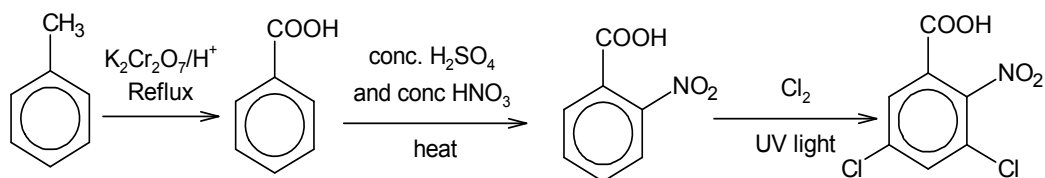
(c) Suggest the structure of **W** and hence identify the organic product **X**.

[2]

(d) Write balanced equations for all the reactions which have occurred.

[3]

- (e) A Chemistry professor wanted his student to devise a synthetic method to convert methylbenzene to 2-nitro-3,5-dichloro acid. His student suggested this method:



The professor identified at least three errors to his suggested method for the synthesis.

- (i) Identify the three errors and explain with reasons.

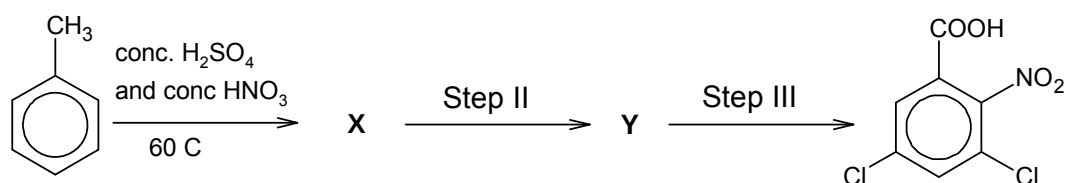
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Given the first step in the synthesis is as follows:



- (ii) State the reagents and conditions for Steps II and III.

Step II:

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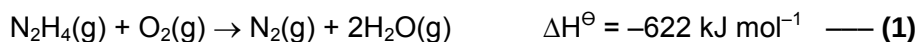
Step III:

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[5]

[Total: 13]

- 3** The major uses of hydrazine, N_2H_4 , are in rocket fuels. Hydrazine was first used as a rocket fuel during World War II for the first rocket-powered fighter plane. These reactions are extremely exothermic and they produce large volumes of hot gas from a small volume of liquid hydrazine.



It has been suggested that hydrazine/fluorine mixtures might be more exothermic than hydrazine/oxygen mixtures.



- (a) (i)** With the aid of a *Data Booklet*, calculate the enthalpy change per mole of hydrazine when it reacts with fluorine.

- (ii)** Suggest a reason for the discrepancy between your calculated value in **(i)** and that given in the question.

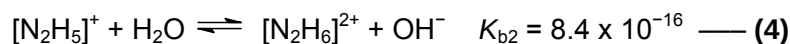
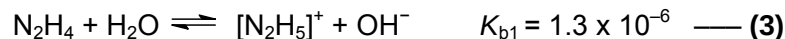
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- (iii)** Explain the difference in $\Delta H^\ominus$ between reactions **(1)** and **(2)** in terms of bond strength.

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.....

[5]

- (b) In some way, hydrazine resembles ammonia in the way it reacts with water. The product from the first reaction can react with a second molecule of water as shown by equation (2).



- (i) Identify one conjugate acid–base pair in equation (3).

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.....

- (ii) Explain why K_{b2} is much smaller than K_{b1} .

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[2]

- (c) Hydrazine also finds many applications in organic chemistry. One such application is the reaction with carbonyl compound to form hydrazone.

- (i) State the type of reaction which occurred when hydrazine reacts with propanone to form the hydrazone.

.....

- (ii) Draw the displayed formula of the organic compound formed.

- (iii) Explain why the organic compound formed from the reaction cannot exist as stereoisomers.

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[3]

- (d) Hydrazine also finds its use in fuel cell to power automobiles and other motorized vehicles. A redox reaction takes place in the cell where hydrazine is oxidised in an alkaline medium to nitrogen gas while oxygen gas is reduced to hydroxide ions at the cathode where reduction takes place.

Write the half-equations which occurred for the redox reaction of hydrazine, and hence the overall reaction of the fuel cell.

oxidation:

.....

reduction:

.....

overall equation:

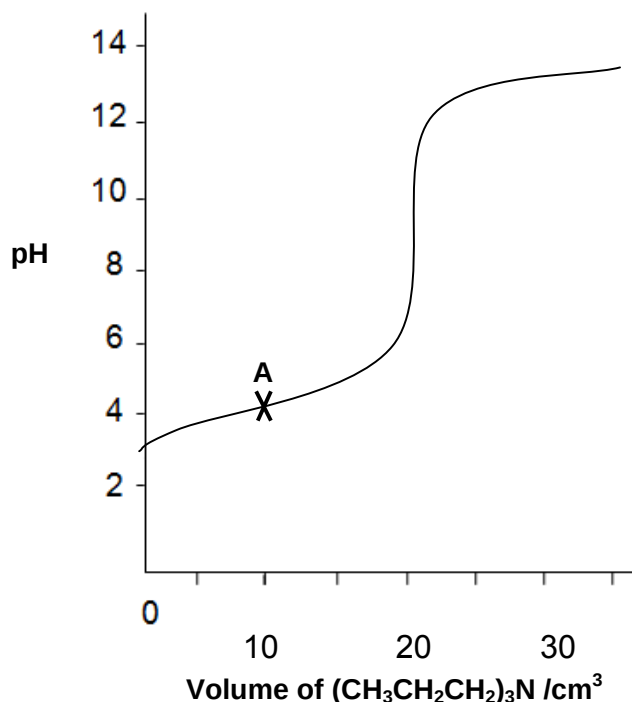
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[3]
[Total: 13]

Section B

Answer any **two** questions on separate answer paper.

- 4(a)** Carboxylic acids are weak acids as they only ionise slightly in water. One example will be the acrylic acid, with the formula of $\text{CH}_2=\text{CHCO}_2\text{H}$. 15.0 cm^3 of 0.30 mol dm^{-3} of acrylic acid is titrated with a strong base, aqueous tripropylamine $(\text{CH}_3\text{CH}_2\text{CH}_2)_3\text{N}$, and its titration curve is as shown below.



- (i) Write a balanced equation for the reaction between acrylic acid and tripropylamine.
- (ii) Calculate the concentration of tripropylamine used in this titration and hence its pH.
- (iii) The solution at point **A** can act as a *buffer*. Explain what is meant by a buffer. Write an equation to show how the solution at point **A** can act as a buffer when a small amount of H^+ is added to it.
- (iv) Suggest with reasoning from the following list, a suitable indicator to be used for the titration.

Indicators	pH range
Methyl violet	0 – 1
Methyl orange	3 – 4
Bromothymol Blue	6 – 7
Phenolphthalein	9 – 10

[6]

- (b) The standard enthalpy change of the above reaction in (a)(i) can be determined experimentally by mixing known volumes of 1.0 mol dm^{-3} acrylic acid and 1.0 mol dm^{-3} tripropylamine. The following results are obtained.

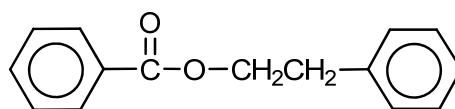
Volume of acrylic acid used = 40.0 cm^3
 Volume of tripropylamine used = 40.0 cm^3
 Initial temperature of mixture = 26°C
 Final temperature of mixture = 32°C

- (i) Use the data given to calculate the standard enthalpy change of the reaction. [Assume specific heat capacity of all solutions = $4.20 \text{ J g}^{-1} \text{ K}^{-1}$].
- (ii) Using the standard enthalpy change calculated in (i), draw a labelled energy profile diagram for the reaction, given that it has an activation energy of $+26 \text{ kJ mol}^{-1}$.
- (iii) The standard enthalpy change of reaction between the NaOH and HCl is found experimentally to be -57 kJ mol^{-1} . Explain if the ΔH° calculated in (i) is plausible.
- (iv) State two possible sources of errors in this experiment.

[7]

- (c) A distinct difference between carboxylic acids and esters is in their odors. Carboxylic acids are noted for their sour, disagreeable odors. On the other hand, esters have sweet pleasant odors often associated with fruits. Hence esters are often used in the food industry as flavoring agents.

A food chemist decides to synthesise the following ester, for possible use as a flavouring agent.



phenethyl benzoate

The **only** organic compound available is phenylethanal ($\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$). Explain how the food chemist is able to synthesise this ester, stating the necessary reagents and conditions for each step.

[2]

- (d) The label on the test tube containing phenethyl benzoate, which is synthesised in (c), came off and it is mixed up with two other compounds with structures as shown below.

- Phenylmethanol, $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
- Phenylethene, $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$

Outline a **sequence of chemical tests** by which you could identify each of the above organic substances. You are **not** allowed to identify the substances by elimination.

[5]

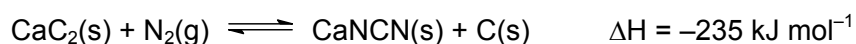
[Total: 20]

- 5(a)** Calcium carbide, CaC_2 , is an ionic compound of calcium and carbon which is one of the most important carbides used in technology. It is used in carbide lamps, in which water drips on the carbide and the acetylene, C_2H_2 , formed is ignited. Calcium hydroxide is also produced in this *equilibrium* reaction.

- (i) Write an equation to show the production of acetylene from calcium carbide.
- (ii) Draw a dot-and-cross diagram to show the arrangement of electrons in acetylene.
- (iii) State the oxidation number of carbon in acetylene and give an example of a carbon-containing species with the same oxidation number.

[4]

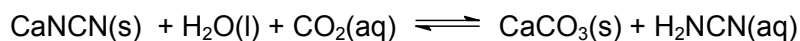
- (b)** Upon heating calcium carbide in nitrogen at 1000°C , calcium cyanamide is produced:



- (i) State the formula of the cyanamide ion and draw its Lewis structure.
- (ii) Using your answer in (i), predict the shape of cyanamide ion and give the hybridisation of carbon atom.
- (iii) Explain qualitatively why the production of calcium cyanamide is carried out at an optimum temperature of 1000°C .

[7]

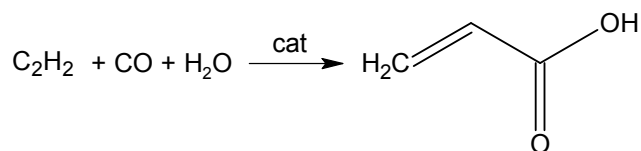
- (c)** Calcium cyanamide is often being hydrolysed to give cyanamide, H_2NCN , which is widely used in agriculture and the production of pharmaceuticals and other organic compounds.



Write an expression for the equilibrium constant, K_c , of the reaction given above. By using the numerical value of this K_c (4.7×10^3) and equilibrium $[\text{CO}_2(\text{aq})]$ of $3.4 \times 10^{-4} \text{ mol dm}^{-3}$, estimate the concentration of H_2NCN that could be produced.

[2]

- (d)** Acetylene can react with carbon monoxide and water, in the presence of a catalyst, to give an acrylic acid. The water molecule loses a proton to form the hydroxyl group.



- (i) Compare and explain the relative acidity of this acrylic acid with 2-chloropropanoic acid, $\text{CH}_3\text{CH}_2(\text{Cl})\text{COOH}$.
- (ii) Acetylene can also react with carbon monoxide and ethanol in a very similar fashion to the above reaction in part (d) to give product **A**. Draw the structural formula of **A**.

- (iii) **A** is then subjected to strong oxidation whereby hot acidified KMnO_4 is used in the reaction.

Draw the structural formula of all the products upon strong oxidation of **A**.

- (iv) Suggest an alternative set of reagent and condition for the oxidation reaction such that there are **only** 2 distinct organic products upon oxidation.

[7]

[Total: 20]

6 This question is about phosphorus and its compounds.

- (a) (i) State the full electronic configuration of phosphorus.
- (ii) Suggest how the first ionisation energy of phosphorus would compare to that of sulfur and explain your reasoning.

[4]

- (b) One of the products from the reaction of phosphorus and sulfur is phosphorus sulfide, P_4S_3 , which is used in small amounts in the tip of a match. On striking a match, this compound burns to give phosphorus(V) oxide, P_4O_{10} and sulfur dioxide, SO_2 .

(i) Construct a balanced equation for the reaction of P_4S_3 to form the respective oxides.

(ii) Both oxides formed in part (b)(i) dissolve in water. Construct an equation for the reaction of each oxide with water.

[3]

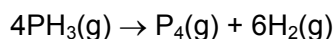
- (c) Phosphorus reacts with limited amount of chlorine to give PCl_3 and excess chlorine to form phosphorus pentachloride, PCl_5 .

(i) Explain why PCl_5 sublimes at 160°C while PCl_3 boils at 76.1°C .

(ii) Describe, with an equation, the reaction of PCl_5 , with water. What would be the effect of adding universal indicator to the product?

[6]

- (d) Phosphine, PH_3 , is a widely used fumigant. The thermal decomposition of phosphine into phosphorus and hydrogen is a *first-order* reaction. The half-life of the reaction is 35.0 seconds at 680°C .



(i) Explain the term in *italics*.

(ii) Sketch a graph of rate against concentration for the decomposition reaction.

(iii) Calculate the rate constant for the reaction.

(iv) Calculate the time taken for 87.5% of the phosphine to decompose.

[7]

[Total: 20]

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